# HIREX™ : THE TRUTH-IN-HIRING PROTOCOL

**Master Architecture Blueprint, Problem Statement Pivot & Technical Implementation Dossier**

**Institution:** JSPM RSCOE, Pune | **Hackathon:** CraftVerse National (PCCOER) | **Team:** Team Dominator | **Version:** 3.2.0 (Pivot Edition)

## 1. Executive Summary & The Problem Statement Pivot

In our previous hackathon iteration, our platform (originally titled *SkillBridge*) was engineered to address broad, macro-level academic-industry curriculum desynchronization. While mathematically robust, that problem statement lacked the intense, visceral operational urgency found in real campus placement offices. The new problem statement (**Placement Hub**) presents an extraordinary opportunity: it narrows the focus to the single most acute crisis in tech recruitment today: **the total collapse of trust in fresher resumes**.

| Parameter | Old Focus (SkillBridge v1.0) | New Focus (HireX Placement Protocol) |
|---|---|---|
| **Core Problem** | College syllabi are outdated compared to Pune/national industry job vacancy skills. | Freshers spray generic resumes with fake claims ('Redis expert'); TPOs forward jobs blindly on WhatsApp; duplicates arrive from multiple sources; recruiters waste 80% of Round 1 interviews on false claims. |
| **Primary User** | Academician / HOD analyzing syllabus gaps. | Three-sided operational network: **Student** (anxious, resume spamming) + **TPO** (overwhelmed by WhatsApp/Excel) + **Recruiter** (exhausted by fake claims). |
| **Resume Handling** | Simple keyword scanner measuring keyword frequency. | **Claim X-Ray (The Liar Detector):** Cross-checks resume claims against real evidence; **blocks unverified claims**; generates truthful 1-page tailored ATS resumes with QR proof. |
| **Opening Ingestion** | Static list of campus placement drives. | **Multi-Source Ingestion & Fuzzy Deduplicator:** Ingests from Partner Portals, WhatsApp Alumni Referrals, and Job APIs; merges duplicates into a single canonical opening. |
| **TPO Capability** | Viewing departmental skill gap analytics. | **Interactive Batch Ranker:** TPO pastes any unexpected JD text and instantly ranks 50+ students by evidence-backed fit score in 200ms with 1-click shortlist export. |
| **Killer Demo Concept** | Curriculum desynchronization percentage curve. | **'The Resume That Can Survive a Technical Round':** Live smartphone scan of resume QR code showing AST code proof + interview grilling defense simulator. |

## 2. The Three-Sided Placement Breakdown (The Problem HireX Solves)

Commercial campus placement platforms in India (e.g., JoinSuperset) automate administrative registration but suffer from a fatal flaw: **they trust unverified uploaded PDFs and student self-claims**. Meanwhile, modern AI resume builders (ChatGPT, Rezi, Teal) accelerate resume fraud by hallucinating buzzwords without proof. HireX establishes the first closed-loop protocol where claims must be proven before being sent to an employer.

**1. THE STUDENT CRISIS: 'The Spray & Pray'**
Facing an 80% ATS rejection rate, students panic and use ChatGPT to inflate resumes with unbacked buzzwords (*'experienced with microservices, proficient in Redis'*). When shortlisted, they are humiliated and exposed within 5 minutes of their first technical interview, destroying their confidence and damaging their college's standing.

**2. THE TPO CRISIS: 'WhatsApp & Excel Hell'**
Job openings trickle in asynchronously via alumni WhatsApp chats, company HR emails, and external web boards. TPOs accidentally broadcast the same opening 3 times under different titles, track applicants on fragmented Google Sheets, and lack any objective way to rank 200 students when a recruiter asks for 20 candidates by tomorrow morning.

**3. THE RECRUITER CRISIS: 'Noise Fatigue'**
Enterprise recruiters receive 800 resumes that all look identical. They cannot distinguish a student who actually architected a multi-threaded system from one who simply copy-pasted the keyword. Engineering teams waste 40+ expensive hours per hire weeding out false positives during Round 1 viva.

# 3. Technical Architecture: The 4-Tier Evidence Ledger

To replace self-claims with mathematical certainty, HireX implements a centralized **Evidence Ledger**. Every claim is assigned to a strict tier based on objective auditability. **Tier 4 claims are strictly blocked from export**.

| Evidence Tier | Verification Source & Methodology | Proof Artifact Generated | Resume Status |
|---|---|---|---|
| **TIER 1: Proctored Assessment** | Proctored algorithmic coding challenge or cognitive neuroscience test session with fullscreen lock and face-tracking telemetry. | Proctoring session ID, timestamped violation log, percentile score, test hash. | **VERIFIED (Full Weight)** |
| **TIER 2: Code Lab / GitHub Audit** | In-browser AST syntax parser analyzing loop depth, recursion trees, cyclomatic complexity + real dependency manifests (requirements.txt, package.json). | SHA-256 cryptographic proof of work, commit cadence timeline, AST cyclomatic grade. | **VERIFIED (Full Weight)** |
| **TIER 3: Evaluated Project / QR** | Mentor-evaluated capstone repository, NPTEL / AICTE course certificate with verifiable public QR signature registry. | Registry credential token, faculty evaluation rubric score, project repository link. | **SUPPORTED (Partial Weight)** |
| **TIER 4: Self-Reported Claim** | Keywords extracted from student resume or profile text with zero corresponding commits, test scores, or certificates. | None. Unbacked textual claim. | **HARD-BLOCKED (Export Denied)** |

# 4. Flagship Feature Specifications (The 6 Pillars of HireX)

Each feature directly eliminates a specific failure mode defined in the problem statement while creating a compelling, interactive live demo for the judges.

## Feature 1: Claim X-Ray (The Fresher Resume Lie Detector)

**What it does:** When a student uploads an existing PDF or pastes resume text, HireX extracts every technical claim and cross-audits it against their personal Evidence Ledger. Each claim is labeled **Green (Verified)**, **Amber (Weak)**, or **Red (Unsupported)**.
**The Hackathon WOW Factor:** When a student uploads a resume claiming *'Experienced in Redis, Microservices & Docker'*, the system immediately highlights Redis in red: *'■ 0 Redis commits, 0 test cases found. Claim BLOCKED from employer submission.'* An inline button lets the student take an instant 2-minute skill quiz to verify the claim on the spot.

## Feature 2: Truthful 1-Page Tailored ATS Resume Studio + Proof QR

**What it does:** Rather than spraying a single generic 2-page resume, students click *'Generate Tailored Resume'* on any target job. HireX synthesizes a clean, 1-page ATS-compliant resume containing **only** the student's verified skills and project achievements matching that specific role.
**The Scan-to-Verify QR Code:** Every generated resume includes a cryptographic QR code in the header. When scanned with a smartphone camera, it opens a public mobile verification page showing the exact AST loop depth, passing pytests, and GitHub commit diffs for every bullet point on that resume.

## Feature 3: Multi-Source Job Ingestion & Fuzzy Deduplication Engine

**What it does:** Ingests openings from three distinct channels: **(1) Campus Partner Drives**, **(2) Alumni WhatsApp/Referral Submissions**, and **(3) External Job APIs**. It normalizes company names, job titles, and locations, then computes skill-set Jaccard similarity to detect duplicate postings.
**Why it helps:** When Barclays posts on their careers portal, an alumnus forwards the link on WhatsApp, and an API scrapes it, HireX merges them into a single canonical card displaying: *'■ Canonical Opening (Merged from 3 Sources): Barclays Careers + Alumni Referral (Kunal S. '23) + LinkedIn API'*.

## Feature 4: Placement Cell (TPO) 'Paste-a-JD' Batch Ranker

**What it does:** Solves the urgent recruiter request crisis. The placement officer simply pastes raw text from any newly received JD, selects a target cohort (e.g., Computer Engineering 2026), and clicks *'Rank Entire Batch'*.
**System Output:** In under 200ms, all 35+ batch students are evaluated using pure deterministic matching (0.7 × required coverage + 0.3 × preferred coverage). The TPO sees an instant leaderboard showing fit scores, confidence bands (e.g. 88% ± 4%), institutional gatekeeping eligibility (CGPA, backlogs), and a 1-click *'Export Verified Shortlist (CSV/PDF)'* button.

## Feature 5: Resume Defence Mode (The Interview Grilling Simulator)

**What it does:** Before attending an interview, the student enters Resume Defence Mode. HireX inspects the candidate's actual GitHub repo facts and simulates tough technical interviewer questions (e.g., *'You claim Redis caching in your ecommerce project. In your repo, cache.py has a fixed 300s TTL. How would you handle cache stampedes if 10,000 requests hit an expired key simultaneously?'*).
**Risk Rating:** Evaluates student answers and flags risk as [Safe], [Shaky], or [Will Be Exposed in Round 1].

## Feature 6: Shared Real-Time Pipeline & Crowdsourced Interview Bank

**What it does:** Replaces WhatsApp chaos with a shared live Kanban pipeline: Applied ➔ TPO Verified ➔ Shortlisted ➔ Technical Round ➔ Offered. Driven by our real-time SSE event broker, when a recruiter or TPO updates an application, the student's screen updates instantly. Furthermore, students who complete interviews submit quick post-round debriefs, building a company-wise repository of real interview questions (Barclays, Persistent, TCS, etc.).

# 5. Screen-by-Screen UI Implementation Map (Code Architecture)

This mapping outlines exactly which files in our existing repository house each capability, ensuring clean modular separation without regressions.

| Platform View / File | User Role | UI Placement & Components | Underlying Service / Engine |
|---|---|---|---|
| **Opportunities View**<br>**OpportunitiesView.tsx** | Student | • Multi-source origin badges & merged duplicate chips on cards.<br>• Alumni Referral drawer ('Request Referral Packet').<br>• Tab 2: **Truthful Resume Studio & Claim X-Ray**.<br>• Tab 3: **Resume Defence Simulator**. | job_deduplicator.py<br>resume_scanner.py<br>jd_skill_analysis.py |
| **Placement Cell Hub**<br>**AcademicianPortalView.tsx** | TPO / Placement Officer | • New Tab: **TPO 'Paste-a-JD' Batch Ranker** with raw JD paste input, batch leaderboard, and CSV export.<br>• New Tab: **Shared Live Application Pipeline** (Kanban).<br>• Company-wise Interview Question Bank. | jd_skill_analysis.py<br>sync_service.py<br>canonical_store.py |
| **Skill Passport**<br>**VerifiedSkillPassport.tsx** | Student | • Unified **4-Tier Evidence Ledger** visual matrix.<br>• Real GitHub dependency manifest & commit inspection.<br>• Instant 'Verify Claim' trigger for unbacked skills. | evidence_ledger.py<br>code_auditor.py<br>skill_intelligence.py |
| **Public Proof Dossier**<br>**PublicVerifyView.tsx** | Recruiter / Public (QR Scan) | • Standalone mobile-friendly public verification route opened by scanning resume QR code.<br>• Displays verified bullet-by-bullet proof, AST cyclomatic depth, and test scores. | canonical_store.py<br>(Public endpoint without authentication wall) |

# 6. The Winning 4-Minute Hackathon Pitch Script

A perfectly choreographed live demo narrative designed to hook technical judges within 30 seconds and secure 1st place.

| Time | Stage Action & Visual Screen | Spoken Pitch Script (Verbatim) |
|---|---|---|
| **0:00 - 0:45**<br>**(The Hook)** | **Screen:** Claim X-Ray in Resume Studio.<br>Upload typical student resume claiming Redis & Microservices. | *'Judges, every fresher in India today is spraying resumes padded with buzzwords they cannot defend. Watch what happens when we upload this resume into HireX Claim X-Ray. Notice that Redis and Kubernetes instantly turn bright RED: 0 commits, 0 test cases found. On any other platform, this fake claim reaches Barclays and wastes 45 minutes of a senior engineer's time. On HireX, unsupported claims are flagged and blocked from employer export.'* |
| **0:45 - 2:00**<br>**(The Truthful Resume)** | **Screen:** 1-Page Tailored Resume Studio with live QR code. | *'Instead of guessing, the student clicks Barclays Full-Stack Drive. HireX synthesizes a clean, 1-page tailored ATS resume where EVERY bullet point has an immutable evidence citation. Judges, please point your phone cameras at this QR code right now. You are seeing live AST code cyclomatic depth, passing pytests, and GitHub commit diffs on your personal phones. HireX produces the only resume that can survive a technical round.'* |
| **2:00 - 3:00**<br>**(The TPO Superpower)** | **Screen:** TPO Portal ➔ Paste-a-JD Batch Ranker.<br>Paste raw text JD ➔ Click 'Rank Batch'. | *'Now let's switch to the College Placement Office. Barclays calls the TPO: "We need 20 backend candidates by 9 AM." Instead of frantic WhatsApp chats and Excel spreadsheets, the TPO pastes the raw JD text into HireX. In 200 milliseconds, our deterministic matcher evaluates the entire 2026 batch. The TPO gets an evidence-ranked leaderboard with confidence bands and 1-click shortlist export.'* |
| **3:00 - 4:00**<br>**(The Close & Moat)** | **Screen:** Multi-Source Merged Cards + Resume Defence Grilling Mode. | *'Notice how our deduplication engine collapsed 3 duplicate WhatsApp, alumni, and API postings into one canonical card. And before stepping into the interview, the student practices in Resume Defence Mode, where AI grills them on their own repo facts. HireX turns campus hiring from a game of resume inflation into a verified meritocracy. Thank you.'* |

# 7. Phased Implementation Waves (The Path to Done)

We execute in 4 clean, non-breaking waves, maintaining our 61-test regression safety net at each step.

| Wave | Core Deliverables | Verification Criteria |
|---|---|---|

| | | |
|---|---|---|
| **Wave 1: Evidence Ledger & Real Proof** | • Implement 4-tier Evidence Ledger data model (Tiers 1–4).<br>• Real GitHub manifest checking (requirements.txt, package.json) so 'Redis' only counts if implemented in code.<br>• Unified taxonomy with strict UNKNOWN fallback. | Unit tests verifying that skills with 0 commits report UNSUPPORTED, while skills with AST commits report TIER_2_VERIFIED. |
| **Wave 2: Job Deduplicator & Ingestion** | • Ingestion pipeline tagging sources: Partner, Alumni WhatsApp, Job API.<br>• Fuzzy deduplication engine using normalized blocking keys + skill Jaccard similarity.<br>• Merged canonical opening card with sources[] provenance chips. | Test cases proving duplicate postings with varying titles collapse into 1 canonical opening with multiple sources. |
| **Wave 3: Claim X-Ray & 1-Page Resume** | • Claim X-Ray parser auditing resume text against evidence ledger (Green/Amber/Red).<br>• Hard-block guardrail preventing export of unbacked claims.<br>• Truthful 1-page ATS Tailored Resume Builder with live Scan-to-Verify QR code and public mobile verification route. | End-to-end verification of resume upload claim detection + responsive 1-page print CSS layout + public QR URL route. |
| **Wave 4: TPO Ranker & Resume Defence** | • TPO 'Paste-a-JD' batch ranker evaluating 35+ students in under 300ms.<br>• Shared live application Kanban pipeline with real-time SSE updates.<br>• Resume Defence Mode interview grilling simulator + company interview debrief bank. | Batch ranking benchmark under 300ms + live Kanban event broadcast test + complete end-to-end user journey walkthrough. |